

$(\mathbb{Z} \times \mathbb{Z})$
 $aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\langle a, b \mid aba^{-1}b^{-1} = e \rangle$
 $\mathbb{Z} \times \mathbb{Z}$
 $S^1 \vee S^1$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$

[illegible]

$S:$

RIP:
0-cell: fm
1-cell: (x)
2-cell: Iy

a/a^z
 $\underline{a=1}$

$T_1(1)$

$K \sim S$

$ab^{-1} = 0.$
 $(ab=b^{-1})$
 $(\pi * 2)/a$
 $(\pi * 2)$

[illegible]

$\frac{1}{2} \frac{dx}{dt} = \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{dx}{dt} \right)$